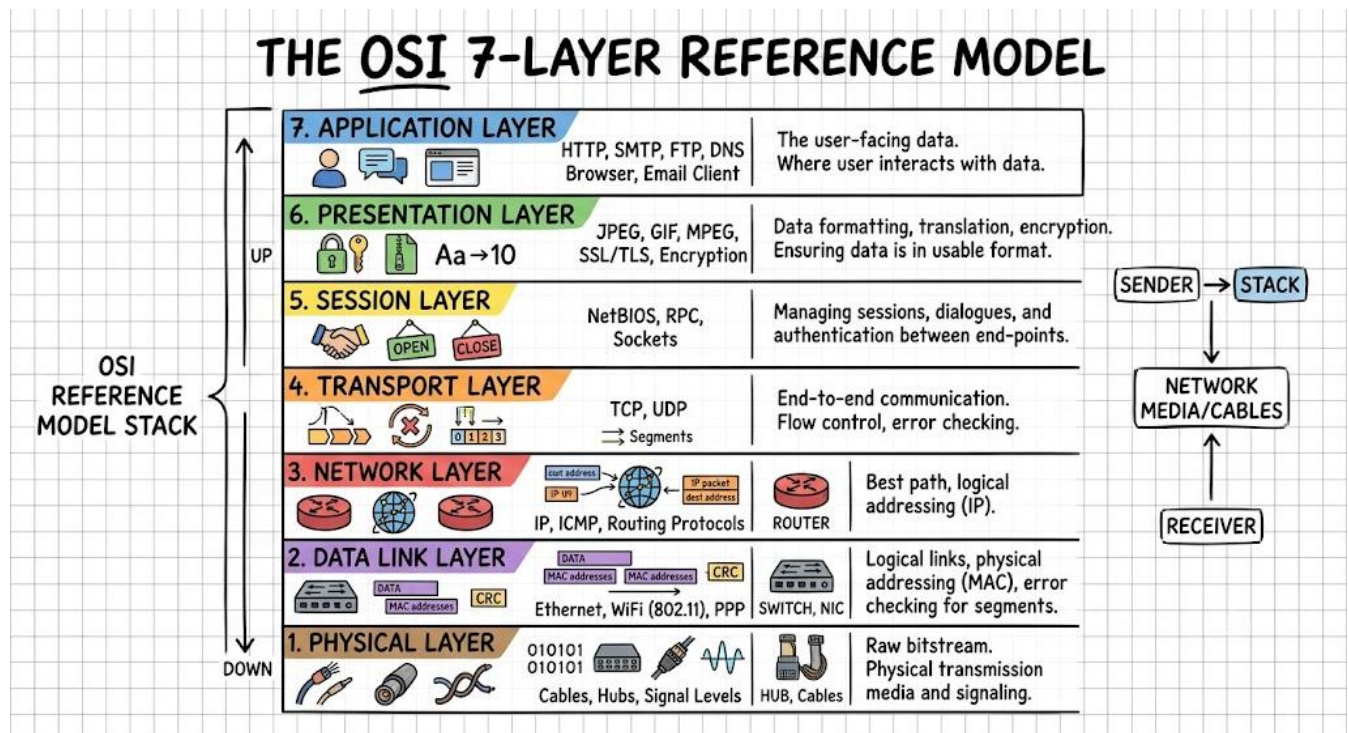


TASK-1



TASK-2

Answers:

- a) Encrypting data before sending: Presentation Layer (Layer 6)
- b) Routing packets across networks: Network Layer (Layer 3)
- c) Converting data to electrical signals: Physical Layer (Layer 1)
- d) Opening a web page in your browser: Application Layer (Layer 7)
- e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

TASK-3

Example App: WhatsApp

When I send a message on **WhatsApp**, multiple OSI layers work together:

1. **Application Layer (Layer 7):** WhatsApp creates and sends the message using the app.
2. **Presentation Layer (Layer 6):** The message is encrypted using end-to-end encryption before it is sent.
3. **Transport Layer (Layer 4):** TCP ensures the message is delivered reliably and in the correct order.

4. **Network Layer (Layer 3):** IP and routers find the best path to deliver the message to the recipient.
5. **Data Link Layer (Layer 2):** The Wi-Fi adapter or mobile network sends data frames to the next device.
6. **Physical Layer (Layer 1):** The data is transmitted as electrical, radio, or optical signals through Wi-Fi, mobile towers, or cables.

This shows how different OSI layers work together to deliver a WhatsApp message quickly, securely, and reliably.

TASK-4

The problem is most likely at the **Application Layer (Layer 7)** or **Presentation Layer (Layer 6)**.

Reason:

- Since you can **ping the website's IP address**, the **Network Layer (Layer 3)** is working correctly.
- If the website does not open, the issue may be with the **Application Layer (HTTP/HTTPS, web server, browser)** or the **Presentation Layer (SSL/TLS encryption certificate)**.
- It could also be caused by a browser problem or the web server not responding to HTTP/HTTPS requests.

Final Answer:

- **Application Layer (Layer 7):** HTTP/HTTPS or browser issue.
- **Presentation Layer (Layer 6):** SSL/TLS encryption or certificate problem.